

ZeyuGuo

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Education

Huazhong University of Science and Technology (985 project)

MPhil candidate in Information and Communication Engineering. GPA: 3.73/4.0.

Sept 2022 – June 2025

Wuhan,China

Wuhan University of Technology (211 project)

B.E. in Electronic Information Engineering. GPA: 3.77/4.0; rank: 13/240, top 6%.

Sept 2018 – June 2022

Wuhan,China

Projects

A Survey on LUT-based Deep Neural Networks Implemented in FPGAs

Oct 2024 - Feb 2025

Description: Given that DSP-based deep neural networks (DNNs) and XNOR-based binary neural networks do not fully utilize the computational advantages of FPGAs, exploring LUT-based implementations for higher-performance DNNs is a meaningful research direction. This survey provides a detailed review of LUT-based DNNs, tracing their evolution and technological pathways while reproducing experimental comparisons of their performance. Finally, potential future research directions are outlined to further advance this field.

An FPGA CI/CD Pipeline.

Sept 2024 - Oct 2024

Description: In this project, I developed an FPGA CI/CD pipeline for Vivado 2024.01. I began by creating a Dockerfile to build a custom Vivado Docker image. The toolchain I implemented automatically initiates a Docker container on a Linux machine whenever a new commit is detected on my Bitbucket repository. Using custom TCL scripts, the pipeline synthesizes and implements the committed FPGA design. Upon completion, the generated bitstream is tagged with the commit hash and securely stored in Artifactory for future reference.

An FPGA-based PVT-Insensitive Picosecond Resolution Timing Generator for ATE.

May 2023 - Mar 2025

Description: In this project, I developed a timing generator (TG) to produce precise timing signals for automatic test equipment (ATE) used in digital IC testing. I introduced an interpolation method to enhance both resolution and linearity, along with a robust offset canceller to improve resilience to PVT (process, voltage, and temperature) variations. The TG was implemented on a Xilinx xcku060ffva1156-2-i FPGA with a 266.5 MHz clock frequency. Test results demonstrate that the proposed method improved timing resolution from 66 ps to 5 ps and reduced the INL (integral non-linearity) from -17.7/+116.3 ps to -2.8/+1.6 ps. Under worst-case PVT conditions, the INL was further refined to -4.7/+4.6 ps. Resource utilization per TG channel was efficient, requiring only 2.9 % of Slices and 1.4 % of BRAM resources.

A High Precision Time-to-Digital Converter (TDC) based on FPGA.

May 2023 - Jan 2024

Description: I developed a time-to-digital converter (TDC) for precise timing measurements in TG generation, using a multi-chain measurement averaging method to overcome intrinsic cell delay limitations and enhance resolution and RMS precision. Bin widths were verified via code density testing, and the TDC was implemented on a Xilinx xcku060ffva1156-2-i FPGA at a 400 MHz clock frequency. Results showed improved average resolution from 28 ps to 1.8 ps and RMS precision from 14.1 ps to 5.4 ps, with a worst-case RMS precision of 7 ps under all PVT corners. Resource utilization per TDC channel was efficient, requiring 3.7 % of CARRY8 and 3.9 % of LUT resources.

Publications

- **Guo Z.** A Survey on LUT-based Deep Neural Networks Implemented in FPGAs. *arXiv*. June 2025.
- **Guo Z,** et al. FPGA-based process, voltage, and temperature insensitive picosecond resolution timing generators with offset correction for automatic test equipment. *Rev. Sci. Instrum.* 1 February 2025; 96 (2): 024702.
- **Guo Z,** et al. A dust sensor monitoring system using Wi-Fi mesh network. *Journal of Physics: Conference Series*. IOP Publishing, 2021, 1754(1): 012129.
- Dust monitoring and processing system based on Wi-Fi Mesh. CN113543058A.

Internship Experiences

FPGA Internship Engineer, Hangzhou SiGauge Technology, Beijing, China

Apr 2024 – Mar 2025

Description: Responsible for development of FPGA based timing generators and pattern streaming. Explored how to minimize the output jitter of timing generators and calibrate the channels of ATE.

Honors & Awards

National Undergraduate Electronics Design Contest: **National Second Prize & Provincial First Prize.**

National Undergraduate FPGA Innovation Design Competition: **National Second Prize.**

Wuhan University of Technology & Huazhong University of Science and Technology: **First Prize Scholarship.**

Wuhan University of Technology: **Merit Student (three times) & Outstanding Graduate.**

Skills

Programming Languages: Verilog HDL, VHDL, System Verilog, C/C++, Python, MATLAB, Tcl, Git, Shell.

Software Tools: Docker, Jenkins, Vivado, Vivado HLS, Quartus ii, Modelsim, MDK5, Altium Designer.

Languages

Mandarin (Native), English (IELTS-7.0).